Quantification: 872_wt_hp_z06
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	11
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1536.09	1363	742.985	px^2
major_axis_length_px	lacuna shape	75.8641	80.3055	22.6571	px
minor_axis_length_px	lacuna shape	26.5145	22.6988	7.8713	px
aspect_ratio	lacuna shape	2.9504	3.0775	0.7901	unitless
eccentricity	lacuna shape	0.9176	0.9457	0.0765	unitless
solidity	lacuna shape	0.8516	0.8855	0.0739	unitless
roots_count	headline	10.7273	10	5.1593	count
ring_length_r30_px	headline	463.818	465	129.907	px
ring_length_r60_px	reported	1197.82	1051	304.003	px
owned_length_px	ownership-dependent	3057.05	3061.83	1166.66	px
edge_count	ownership-dependent	190.818	192	83.4803	count
mean_edge_length_px	ownership-dependent	16.4087	15.9412	1.2798	px
perimeter_px	normalised	201.516	212.409	59.5161	px
ring_area_r30_px2	normalised	8159.55	8564	1316.54	px^2
ring_area_r60_px2	normalised	21087	22238	2804.29	px^2
in_frame_fraction_r30	normalised	0.9903	1	0.0323	unitless
in_frame_fraction_r60	normalised	0.9827	1	0.0574	unitless
ring_density_r30	normalised	0.0569644	0.0567477	0.0126818	px^-1
ring_density_r60	normalised	0.0576194	0.0568671	0.0151487	px^-1
roots_per_100px_perimeter	normalised	5.2797	5.8974	2.0167	per 100 px
ring_attached_length_r30_px	network v2	432.727	436	132.617	px
ring_attached_length_r60_px	network v2	1039.45	849	387.023	px
ring_length_w_r30_px	network v2	526.04	528.262	152.028	px
ring_length_w_r60_px	network v2	1362.03	1219.37	350.704	px
sholl_crossings_r10	network v2	10.1818	10	3.6005	count
sholl_crossings_r20	network v2	13	12	4.4497	count
sholl_crossings_r30	network v2	15.2727	16	3.952	count
loop_count	loops, report only	26.5455	23	17.3284	count
loop_edge_share	loops, report only	0.4922	0.4625	0.1201	share
ring_width_mean_r30_px	width	7.3259	7.3404	0.5845	px

Per field
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measure	value	units
analysed_area_px2	1.03101e+06	px^2
total_skeleton_length_px	39292	px
canalicular_length_density_per_px	0.0381102	px^-1
junction_count	1318	count
junction_density_per_px2	0.00127836	px^-2
lacuna_count	12	count

lacunae_per_px2	1.164e-05	px^-2
skeleton_component_count	153	count
mean_component_length_px	256.81	px
median_component_length_px	17	px
field_length_density_w_per_px	0.0432185	px^-1
field_density_without_flagged_per_px	0.0494838	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	10	px
unattached_component_count	1	count
canalicular_length_density_attached_per_px	0.0381102	px^-1
field_length_density_with_unattached_per_px	0.0381199	px^-1
loop_count	391	count
loop_edge_share	0.5501	share
edge_count	2623	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6256	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	37753	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2178	97.2133	31.5882	3.0775	0.9457	0.7666	-1.4507	132.953	694.614	False
2	1532	90.3989	22.6854	3.9849	0.968	0.8855	-1.4786	133.692	263.431	False
3	1458	82.9994	23.987	3.4602	0.9573	0.8423	-1.4301	272.217	418.857	False
4	2267	64.9571	46.3919	1.4002	0.6999	0.896	-1.1815	380.946	282.045	False
5	3165	121.397	33.5624	3.6171	0.961	0.8709	-1.4322	386.945	67.1896	False
6	743	47.9512	20.7766	2.3079	0.9013	0.8888	-1.5064	382.218	839.734	False
7	712	46.6981	20.2555	2.3054	0.901	0.8856	-1.5028	499.643	510.035	False
8	1363	80.3055	21.9882	3.6522	0.9618	0.8961	-1.3459	561.869	697.237	False
9	669	39.9711	24.8769	1.6068	0.7827	0.7707	-1.1357	585.831	14.0269	True
10	1360	67.805	25.7054	2.6378	0.9254	0.9484	-1.4973	614.482	323.551	False
11	846	53.8777	22.02	2.4468	0.9127	0.8004	1.5387	642.791	211.617	False
12	1273	80.9015	22.6988	3.5641	0.9598	0.6866	-1.5697	768.217	476.052	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	10	465	967	3209.31	211	15.21	7.8301
2	12	356	827	2856.43	176	16.2297	6.8469
3	20	568	1572	3825.89	240	15.9412	7.7049
4	13	578	1524	2231.61	140	15.9401	8.0859
5	18	741	1762	5347.95	367	14.5721	7.3404
6	8	370	1051	3398.73	192	17.7017	7.2474
7	4	309	986	1540.44	89	17.3083	6.4548

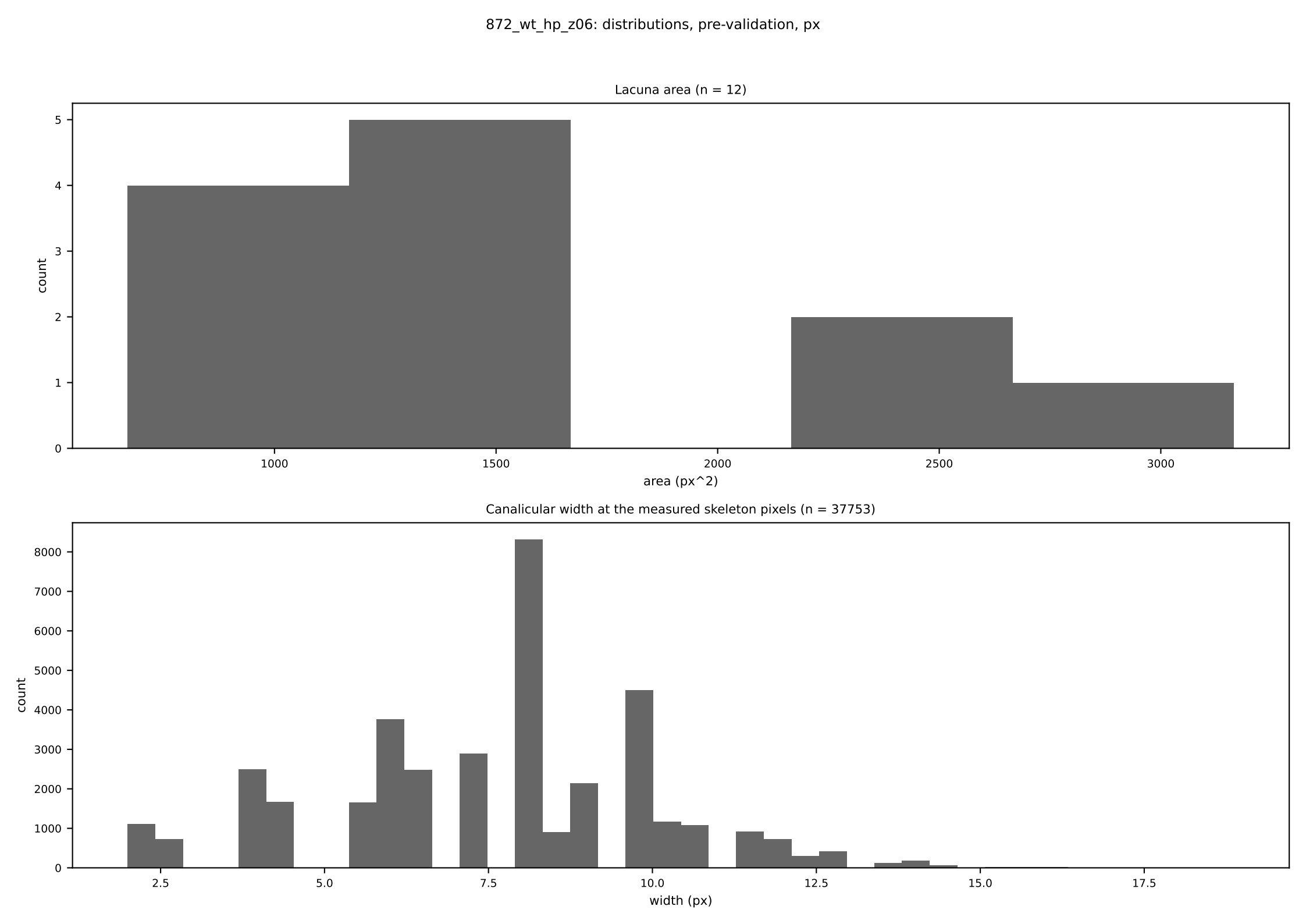
8	12	476	1204	2866.19	157	18.256	6.958
9	8	353	919	2462.03	161	15.2921	7.9105
10	3	396	1031	1189.72	65	18.3034	6.5111
11	9	509	1305	4099.43	266	15.4114	8.1131
12	9	334	947	3061.83	196	15.6216	7.4919

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	292.451	10015	25442	1	1	0.0464304	0.038008	3.4194
2	218.853	8881	23379	1	1	0.0400856	0.0353736	5.4831
3	215.095	8592	22238	1	1	0.066108	0.0706898	9.2982
4	212.409	8564	22184	1	1	0.0674918	0.0686982	6.1203
5	301.179	9919	22490	0.893	0.8096	0.0747051	0.0783459	5.9765
6	123.154	6283	18209	1	1	0.0588891	0.0577187	6.4959
7	119.983	6180	18000	1	1	0.05	0.0547778	3.3338
8	196.024	8388	22420	1	1	0.0567477	0.0537021	6.1217
9	129.29	4152	11045	0.6578	0.6072	0.0850193	0.0832051	6.1876
10	167.539	7464	18130	1	1	0.0530547	0.0568671	1.7906
11	152.61	6844	16739	1	1	0.0743717	0.0779616	5.8974
12	217.38	8625	22726	1	1	0.0387246	0.0416703	4.1402

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	456	902	528.262	1086.45	10	12	10
2	354	733	409.877	944.796	7	12	11
3	562	1551	639.203	1781.84	13	18	16
4	531	1420	661.973	1752.52	11	16	21
5	702	1735	850.754	2008.66	19	22	21
6	330	745	416.806	1219.37	10	9	16
7	244	584	340.208	1117.06	6	7	15
8	436	849	543.961	1362.57	11	15	19
9	347	887	409.806	1051.35	6	11	10
10	347	798	441.463	1160.82	8	10	12
11	484	1288	582.203	1494.06	10	13	16
12	314	829	371.735	1054.18	7	9	11


872_wt_hp_z06: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
1000
1500
2000
2500
3000
Canalicular width at the measured skeleton pixels (n = 37753)
count
width (px)
0
1000
2000
3000
4000
5000
6000
7000
8000
2.5
5.0
7.5
10.0
12.5
15.0
17.5